A PROJECT TITLE

ON

Hospital Re-admission Prediction Using Elastic Net Regularized Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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1. **Abstract:**

Hospital re-admission is a major challenge in healthcare because many patients are admitted to the hospital again soon after being discharged. Frequent readmissions increase healthcare costs, place extra pressure on hospitals, and may affect the quality of patient care. Predicting re-admissions in advance can help doctors provide better treatment and reduce unnecessary hospital visits.

This project proposes a Hospital Re-admission Prediction system using Elastic Net Regularized Logistic Regression**.** Elastic Net combines the advantages of Lasso and Ridge regularization, making the model more accurate and effective. It also helps in selecting the most important features while reducing the impact of unnecessary or highly correlated data.

The model uses patient information such as age, gender, medical history, diagnosis, laboratory test results, previous admissions, length of hospital stay, and medications. Before training the model, the dataset is preprocessed by handling missing values, converting categorical data into numerical form, and scaling the features to improve prediction performance.

The trained model predicts whether a patient is likely to be readmitted after discharge. Its performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and AUC-ROC to measure the quality of predictions.

The proposed system helps healthcare providers identify patients who are at high risk of re-admission so that they can provide proper follow-up care, additional monitoring, and personalized treatment plans. This can improve patient outcomes, reduce hospital readmission rates, and lower healthcare costs. Overall, the project demonstrates that Elastic Net Regularized Logistic Regression is a simple, reliable, and effective machine learning approach for predicting hospital re-admissions and supporting better clinical decision-making.

2. INTRODUCTION:

Hospital re-admission occurs when a patient is admitted to the hospital again within a short period after discharge. Frequent readmissions increase healthcare costs, reduce hospital efficiency, and may indicate the need for better patient care. Predicting patients who are at risk of readmission helps hospitals provide timely treatment and improve patient outcomes.

This project uses Elastic Net Regularized Logistic Regression to predict hospital re-admissions. The model analyzes patient information such as age, medical history, diagnosis, previous admissions, laboratory test results, and length of hospital stay. Elastic Net combines the advantages of Lasso and Ridge regularization, helping to select important features and improve prediction accuracy.

The proposed system assists healthcare professionals in identifying high-risk patients before discharge, enabling better follow-up care and reducing unnecessary re-admissions. Overall, this project demonstrates how machine learning can improve healthcare decision-making and enhance the quality of patient care.

3. SOFTWARE REQUIREMENTS:

Software	**Requirement**
Operating System	Windows 10/11 or Linux
Programming Language	Python 3.x
IDE	Jupyter Notebook / Google Colab / VS Code
Libraries	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
Dataset	Hospital Re-admission Dataset (CSV)

Mrs. I V S L Haritha

Signature of the Guide **Course Instructor**